

JEE MAIN REVISION MASTER: QUADRATIC EQUATIONS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Quadratic Equations - Core Application	3 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Quadratic Equations - Core Application (Moderate Yield — 3 PYQs)

Core Principle: Key problem pattern in Kcet Quadratic Equations.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "If AM and GM of roots of a quadratic equation are 5 and 4, respectively, then the quadratic equation is..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "Given that a, b and x are real numbers and $a < b, x < 0$, then..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "If α and β are roots of the equation $x^2 = x + 1 = 0$, then $\alpha^2 + \beta^2$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

If AM and GM of roots of a quadratic equation are 5 and 4, respectively, then the quadratic equation is

- (A) $x^2 - 10x - 16 = 0$
- (B) $x^2 + 10x + 16 = 0$
- (C) $x^2 + 10x - 16 = 0$
- (D) $x^2 - 10x + 16 = 0$

Correct Answer: None

Solution Analysis: Let a and b be the roots the quadratic equation. Then, the quadratic equation is $x^2 - (a+b)x + ab = 0$ (i)
It is given that AM=5 and GM=4 i.e. $(a+b)/2 = 5$ and $\sqrt{ab} = 4 \Rightarrow a+b=10$ and $ab=16$ On substituting these values in Eq. (i), we obtain. Hence, $x^2 - 10x + 16 = 0$ is the required equation.